

37 Surplus & surplus management

6 primary questions | Oldest to newest

2026 objectives: 5.1.1, 5.2.1, 5.2.3-5.2.6

4 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P2 April 2005 Q2	9 marks
CA1-P2 September 2009 Q4	13 marks
CA1-P2 April 2011 Q7	20 marks
CA1-P2 April 2018 Q3	9 marks
CP1-P2 September 2019 Q1	40 marks
CP1-P2 April 2024 Q2	50 marks

CA1-P2 | April 2005 | Q2

[Current topic](#) | [9 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 2**
- (i) State the reasons why the surplus/deficit arising between successive valuations of a benefit scheme would be analysed. [4]
 - (ii) Describe the steps that would be followed to perform such an analysis. [5]
- [Total 9]

CA1-P2 | September 2009 | Q4

[Current topic](#) | 13 marks total | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** A large company operates a defined benefits scheme for its employees. A supervisory valuation of the scheme has just been completed. The supervisory valuation must be carried out on a set of methods and assumptions as specified by the regulatory authority. The regulatory authority has recently introduced a new set of methods and assumptions, and this is the first valuation carried out on the new set of methods and assumptions.

The results of the valuation show that the accrued liabilities are greater than existing assets giving a deficit, whereas the previous supervisory valuation showed a large surplus.

Describe the various factors and events that could explain this change from surplus to deficit. [13]

CA1-P2 | April 2011 | Q7

Current topic | 20 marks total | Original PDF p4

Related: Ch 18, Ch 3.

- 7** A company which previously sold with profits life assurance directly through employed agents is now closed to new business. The company is collecting data in order to investigate its mortality experience.

(i) Explain why ideal data for this purpose may not be available. [2]

(ii) Discuss how the company can use the limited data it has collected. [4]

The company has publicly stated its commitment to treating its customers fairly.

(iii) Discuss why the company has made this commitment. [5]

In recent years bonus levels on the remaining in-force policies have fallen. Some policyholders have written to the company in response to these falls, to complain that they have not been treated fairly.

(iv) Outline the points the company may make in response to these complaints. [9]

[Total 20]

CA1-P2 | April 2018 | Q3

Current topic | 9 marks total | Original PDF p2

Examiner report and original paper: use the links for this question in [index.html](#).

- 3** A life insurance company has historically written individual annuity business. The company has recently closed to new business and has promised shareholders that this will lead to an increase in its profits over the next three years.
- (i) Describe the issues the insurance company needs to consider when projecting its profits over the next three years. [4]
- (ii) Discuss the management actions that could increase the profits for the company over the next three years. [5]
- [Total 9]

CP1-P2 | September 2019 | Q1

Current topic | 40 marks total | Original PDF p2

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 30, Ch 26, Ch 34.

1 Background

XYZ is a general insurance company in a developed country, specialising in motor insurance. It has been one of the market leaders by market share for ten years. However, it has not made a profit in the last two years.

An overseas insurance company, ABC Corporation ('ABC'), is looking to purchase XYZ and has been provided with the following information:

Motor insurance results

An initial analysis of XYZ's claims experience shows that overall, claim numbers over the last two years have increased by 20%. In particular, there has been a significant increase in claims connected to car and motorcycle accidents.

Company retirement benefit schemes

XYZ has two retirement benefit schemes: a defined benefit scheme and a defined contribution scheme. Five years ago the defined benefit scheme closed to future service. Members of that scheme joined the defined contribution scheme in respect of future service.

Three years ago, XYZ agreed to make annual payments into the defined benefit scheme for the following ten years to eliminate the funding deficit. However, recent results suggest that the deficit has increased, indicating that XYZ may be required to increase its payments into the defined benefit scheme.

The defined contribution scheme is a with-profits policy managed by a life insurance company which provides investment and administration services. The policy guarantees a minimum investment return of 3% p.a. on funds invested before retirement. XYZ matches employee contributions into the scheme up to a maximum of 5% of salary.

Other information

You have also been provided with the following information:

- XYZ ran a promotional campaign last year to increase the number of motor insurance policies sold to young people between the ages of 18 and 24.
- From next year, the government is introducing legislation which will require all pension scheme funding deficits to be eliminated within five years.

CP1-P2 | September 2019 | Q1 - continued

Current topic | 40 marks total | Original PDF p3

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 30, Ch 26, Ch 34.

Questions

- (i) Set out the reasons why ABC may wish to purchase XYZ. [5]
 - (ii) Describe, using examples, how XYZ can manage and control its risks in connection with its motor insurance policies. [10]
 - (iii) Set out possible reasons for the worsening of the funding position in the defined benefit scheme. [10]
 - (iv) Discuss potential options available to improve the funding position of the defined benefit scheme. [8]
 - (v) Discuss the key risks of the defined contribution pension scheme. In your answer you should comment on each risk from the perspective of XYZ and the scheme members. [7]
- [Total 40]

CP1-P2 | April 2024 | Q2

Current topic | 50 marks total | Original PDF p3

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 15, Ch 32, Ch 38, Ch 2.

- 2** A small life insurance company (Company X) writes only immediate annuities. An extract from its latest financial information is as follows:

<u>Regulatory balance sheet at end of year</u>	£m
Assets at fair value	<u>100</u>
Liabilities	
At fair value	(85)
Required solvency and capital margins	<u>(10)</u>
	<u>(95)</u>
Free assets	<u>5</u>

New business is typically issued based on expected returns close to fair value investment returns. During the last financial year new business was priced assuming an average investment return assumption of 5% p.a. The end of year fair value investment return assumption for the company's liabilities is 7% p.a.

Company X has had a full independent review of its data and no issues were found. However, the company's annual profits have been declining for several years and the company is considering two options to increase its annual profits in the future. The two options are:

Plan A: Increase the profitability of its immediate annuity business.

Plan B: Increase overall company profits by selling different insurance products, possibly under a different brand name.

- (i) Discuss Company X's likely investment strategy for the assets backing its immediate annuity business. [7]
- (ii) Set out how the investment return assumption for the fair value of liabilities would be determined. [2]
- (iii) Explain why the investment return assumption for the fair value of liabilities can be higher than the average investment return assumption used in pricing last year's new business. [4]
- (iv) Describe how an analysis of experience could be used to investigate Plan A. [7]
- (v) Discuss a range of possible actions Company X could take to increase its profits under Plan A. [15]

CP1-P2 | April 2024 | Q2 - continued

[Current topic](#) | [50 marks total](#) | [Original PDF](#) p4

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 15, Ch 32, Ch 38, Ch 2.

- (vi) Discuss the factors Company X should consider when deciding if it should sell insurance under a different brand name under Plan B. [5]
 - (vii) Discuss the factors Company X should consider when deciding if it should adopt Plan A or Plan B. [10]
- [Total 50]